

Taiwan Fair Trade Commission

With rapid advancements of Internet technologies, there has been a surge of innovative business models, for example, e-commerce, cloud-computing, use of big data, the Internet of Things and sharing economies. However, the emergence of new business models may bring about further competition concerns. Taking on an important role, competition agencies need to advance their understanding of relevant markets and competitive behavior in business and equip themselves with up-to-date analytical tools and law enforcement capabilities, in order to ensure and promote market competition. In the context of increasingly complex markets, the Taiwan Fair Trade Commission (the ‘TFTC’) collaborated with external experts and economists to develop algorithm-based systems in 2022 that were designed to monitor major daily necessities such as soybean, wheat, corn and eggs. Moreover, the TFTC performed empirical analysis on market definition, implemented merger simulation in physical retail markets, and established geographic information systems. These ‘computational tools’ have been used to facilitate the TFTC’s law enforcement activities.

I. Algorithm-based monitoring and detection systems

By using different methods, including Q-learning algorithms and marketing margin approaches along with web crawler tools, the TFTC developed specific mechanisms to monitor and detect whether pricing practices for soybean, corn, wheat and other daily necessities/bulk commodities were likely to raise concerns over concerted actions:

- With the Q-learning algorithm, the 2022 price and output data in relevant markets was processed and used to test and adjust model parameters. After repeating a large number of rounds and learning with the simulator, the TFTC built a simulation model of vertical and horizontal concerted actions. The analytical results showed that there was lack of evidence indicating the presence of collusion among suppliers of daily necessities/bulk commodities such as soybean, corn and wheat.
- Under the law of one price, the TFTC employed models of VAR and VECM, the Johansen cointegration test and the Granger causality test to analyze marketing margins of daily necessities. The analytical results are summarized as follows:

- Downstream prices of daily necessities including soybean, corn and wheat did not efficiently reflect their respective upstream prices. One possible explanation was that the domestic supply of daily necessities was subject to import regulations on bulk grain.
- Prices of most daily necessities remained relatively stable, but prices for some commodities, for example milk and eggs, often rose faster than they fell. More evidence was required to be carefully scrutinized to determine whether market price changes were resulting from concerted actions.

II. Empirical analysis on market definition and merger simulation in physical retail markets as well as creation of geographic information systems

As the domestic economy has undergone different developmental stages, business environments in retailing, the number of retail businesses and retail market structures have been changing. Drastic changes in market dynamics were observed in the retail sector particularly during the COVID-19 pandemic. To respond to these changes, the TFTC, external experts and economists worked together in a combined effort to develop more practical methods for merger analysis in physical retail markets. The TFTC designed and conducted consumer surveys to collect required market data and utilized economic tools to define relevant markets. The TFTC also made use of geographic information systems ('GIS') displaying the spatial distribution of retail businesses and retailer density. Furthermore, econometric methodologies were applied to emulate competitive effects in post-merger markets:

- Following the hypothetical monopoly test, the TFTC reviewed the data collected from consumer surveys to define a relevant market. The result indicated that hypermarkets across the nation could be defined as a relevant product market.
- On the basis of the 2022 price and output data in the relevant market, the TFTC made use of the logistic regression model to analyze anti-competition effects arising from a proposed merger. The model suggested that the merger was likely to have anti-competitive effects as a result of the level of the increase in post-merger prices.
- The TFTC completed domestic retail store mapping with a GIS system. The mapping result demonstrated that physical retail outlets were highly concentrated in metropolitan areas.

The TFTC employs various computational tools, including R programming, Python, GIS and digital forensic tools for the purpose of data cleaning and warehousing. In competition enforcement activities, the TFTC selects computational tools applicable and appropriate to specific cases. For example, in merger investigations, the TFTC defined relevant markets by means of econometric methodologies or GIS, and then calculated market shares and levels of market concentration. To assess unilateral effects or other anti-competitive effects, the TFTC looked at different indicators including GUPPI and compensating marginal cost reduction analysis. In respect of concerted actions, the TFTC built relevant time series models to examine trends and correlations of prices and costs, thereby identifying signs of potential concerted actions.

A. Hypermarket mergers

The TFTC made attempts with different methods to sort, warehouse and clean a large number of data sets such as tax registrations for businesses and questionnaire survey results. These methods included web crawling, text mining and machine learning in R programming and Python. Regression models and critical loss analysis were implemented to define relevant markets. The analytical process started with the hypermarket sector, i.e. the initial candidate market. The TFTC then calculated market shares and market concentration as well as assessed unilateral effects and other anti-competitive effects.

B. Cinema mergers

A new concept used by the TFTC to define local markets in cinema mergers is isochrone-based catchment areas. First, the length of time or the distance consumers would be willing to travel to reach the locations of merging parties and other cinemas was determined. On account of the diversity of domestic transport modes for personal travel--walking, bicycle, car, bus, metro and rail, the journey time or the distance was decided by the most commonly used transport modes. Isochrones were then constructed with GIS software to measure overlapping catchment areas of individual cinemas, assess the level of competition among cinemas and define geographic markets.

C. Price fixing among ice-making businesses on one offshore island

Tools for extraction of digital evidence and digital forensics were used to acquire instant messaging chats from mobile devices owned by relevant businesses. The TFTC cross-checked these communication histories and the dates of price increases and obtained direct evidence supporting price fixing.

D. Price fixing in carbonated drinks

Data warehousing and cleaning with R was applied to a large number of data sets around carbonated drinks. The impact of monthly changes was taken into consideration based on the interrupted time series approach to build a regression model for different types of carbonated drinks. The TFTC gauged the elasticity of demand by evaluating the differences in outcomes before and after price adjustments. The TFTC further compared the elasticity of demand and the extent of increases in prices to review whether price adjustments were considered economically rational.

III. In 2023, the TFTC will take the following measures to enhance the effectiveness of antitrust enforcement activities and continue technology capacity building in its law enforcement.

- Establishing applicable modules/packages for economic analysis in antitrust: The modules and packages will be tested and verified in real-life situations. The TFTC will be endeavoring to stay on top of the latest information on relevant programming languages and modules/packages for antitrust analysis to optimize the accuracy of its current models/packages.
- Strengthening expertise of digital forensics and capabilities of digital data extraction: Procuring GIS software and password cracker tools with the aim to develop cyber capabilities and reinforce the efficacy of antitrust enforcement.
- Improving mobile forensics skills and implementing new technologies in enforcement activities: To achieve efficiency improvements in antitrust enforcement, procurement of specialized tools for mobile forensics is underway. This is aimed at effectively deterring potential anti-competitive practices by collecting and analyzing relevant evidence in a faster and more precise way.